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$\mathbb{Z}_n$

As we have already learned $\mathbb{Z}/n\mathbb{Z} = \{0, 1, 2, 3, \ldots, n-1\}$. Fortunately, there is a shorter notation to represent this object. That notation is $\mathbb{Z}_n$. For example,

$$\mathbb{Z}_4 = \mathbb{Z}/4\mathbb{Z} = \{0, 1, 2, 3\}$$

$$\mathbb{Z}_7 = \mathbb{Z}/7\mathbb{Z} = \{0, 1, 2, 3, 4, 5, 6\}.$$

Caution: $(\mathbb{Z}/4\mathbb{Z})^\times = \{1, 3\}$ and $\mathbb{Z}/4\mathbb{Z} = \{0, 1, 2, 3\}$ are not the same thing, so you can't write $(\mathbb{Z}/4\mathbb{Z})^\times = \mathbb{Z}_4$. To make things clear, $(\mathbb{Z}/4\mathbb{Z})^\times$ is the set of totatives of 4, while $\mathbb{Z}_4 = \mathbb{Z}/4\mathbb{Z}$ is the set of remainders you can get when dividing by 4.

Example: Some, but not all, of the $\mathbb{Z}_n$'s are fields. For example, $\mathbb{Z}_3 = \{0, 1, 2\}$ with addition $\pmod 3$ and multiplication $\pmod 3$ is a field. Let's see why: (1) It is pretty clear that modular multiplication distributes over modular addition, so the distributive property is taken care of. (2) Now, we have to check that $(\mathbb{Z}_3, + \pmod 3)$ is a commutative group. It contains 0, which is the additive identity. Modular addition is obviously associative and commutative. The additive inverse of 0 is 0, and 1 and 2 are additive inverses of each other. Thus, $(\mathbb{Z}_3, + \pmod 3)$ is a commutative group. (3) Finally, we have to check that $(\mathbb{Z}_3 \setminus \{0\}, \times \pmod 3)$ is a commutative group. It contains 1, which is the multiplicative identity. Modular multiplication is associative and commutative. The only elements in this set are 1 and 2, both of which are self-inverses. Therefore, $(\mathbb{Z}_3 \setminus \{0\}, \times \pmod 3)$ is a group, and $(\mathbb{Z}_3, + \pmod 3), \times \pmod 3)$ is a field.

Non-example: $\mathbb{Z}_4$ is not a field. This is because $\mathbb{Z}_4 \setminus \{0\} = \{1, 2, 3\}$ is a not a group under multiplication $\pmod 4$: the element 2 doesn't have a multiplicative inverse. Make sure you understand why.

Exercises

(1.) Find two $\mathbb{Z}_n$'s that are fields and two that aren't. Can you make an educated guess about which $\mathbb{Z}_n$'s are fields?